

Exam 1: Basics of R – Version C

Exam Agreement

This exam is an **individual assessment** and you should neither offer help to nor receive help from another person.

Cell phones are to be handed in to the instructor and all other electronic devices must be stored in bags at the front of the room.

No smart glasses or other wearable electronic devices are to be powered on during this exam except with accommodations.

You may use **no physical or electronic** resource to complete the work.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Section: _____ **Date:** _____

Include this signed page as the first page of your submitted work.

Exam Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

What are four things that all computers have in common? Describe each one briefly.

Question 2: Command line and syntax

Explain the syntax of a function. What are the three main components and what do they do?

Question 3: Data Frames

In the data frame `beets` (Fig. 1), what is the mean sugar percentage in harvest 1?

beets	30 obs. of 5 variables
\$ harvest:	Factor w/ 2 levels "harv1","harv2": 1 1 :
\$ block :	Factor w/ 3 levels "block1","block2",...:
\$ sow :	Factor w/ 5 levels "sow1","sow2",...: 3 4
\$ yield :	num 128 118 95 131 136 ...
\$ sugpct :	num 17.1 16.9 16.6 17 17 17 17 16.7 16.4

Figure 1: The beets data set.

Question 4: Matrices and Arrays

Create a 5 dimensional array. Each dimension should have a size of at least 3 and the data should be random numbers.

Question 5: Lists

Using the `anoletree` data set (Fig. 2), how would you reference the named number 0.268 in the member maps?

anoletree	List of 6
\$ edge	: num [1:162, 1:2] 83 84 85 86 87 88 89 90 ...
\$ Nnode	: int 81
\$ tip.label	: Named chr [1:82] "ahli" "allogus" "rubrib..."
..- attr(*, "names")=	chr [1:82] "1" "6" "70" "43" ...
\$ edge.length:	num [1:162] 0.268 0.209 0.402 0.826 0.768...
\$ maps	:List of 162
..\$:	Named num 0.268
.. ..- attr(*, "names")=	chr "TG"
\$.. Named num 0.209	

Figure 2: The `anoletree` data set.

Question 6: Class Coercion

To count the number of NA's in the data frame `df`, you can use `sum(is.na(df))`. Explain how this works, including any class coercions that might happen.

Question 7: Data Classes and Special Values

You are trying to reference the yield data in the data set **beets** (Fig. 1) using the following line:

```
beets$Yield
```

R returns NULL. This is clearly not right. Why does it return this and how can you fix it?

Question 8: Factors

Make a vector of ordered factors describing how fluffy each of the following dog breeds: German shepherd dog, pug, poodle, golden retriever, Labrador retriever. Your ordered factor must have at least three levels. Store this vector as **dog_fluffiness**.

Question 9: Vectors

The following is not a valid vector in R. Describe why this is the case.

```
vec <- c(10, matrix(runif(30, nrow = 10), 9))
```

Question 10: Troubleshooting

You are running a friend's code and get the following error:

```
Error in big.m[7, 1:5] : subscript out of bounds
```

What can you tell your friend to help them fix this error?